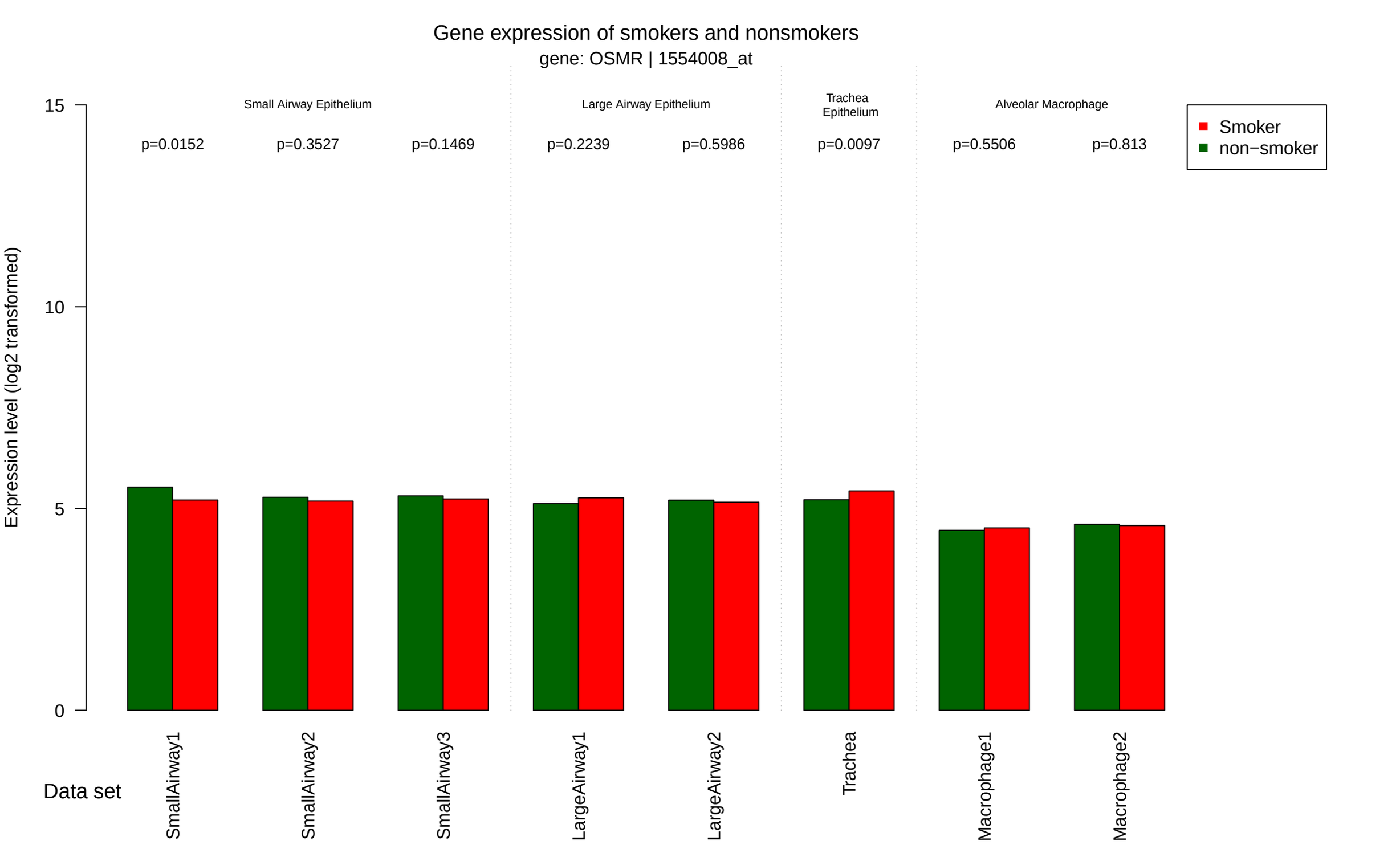
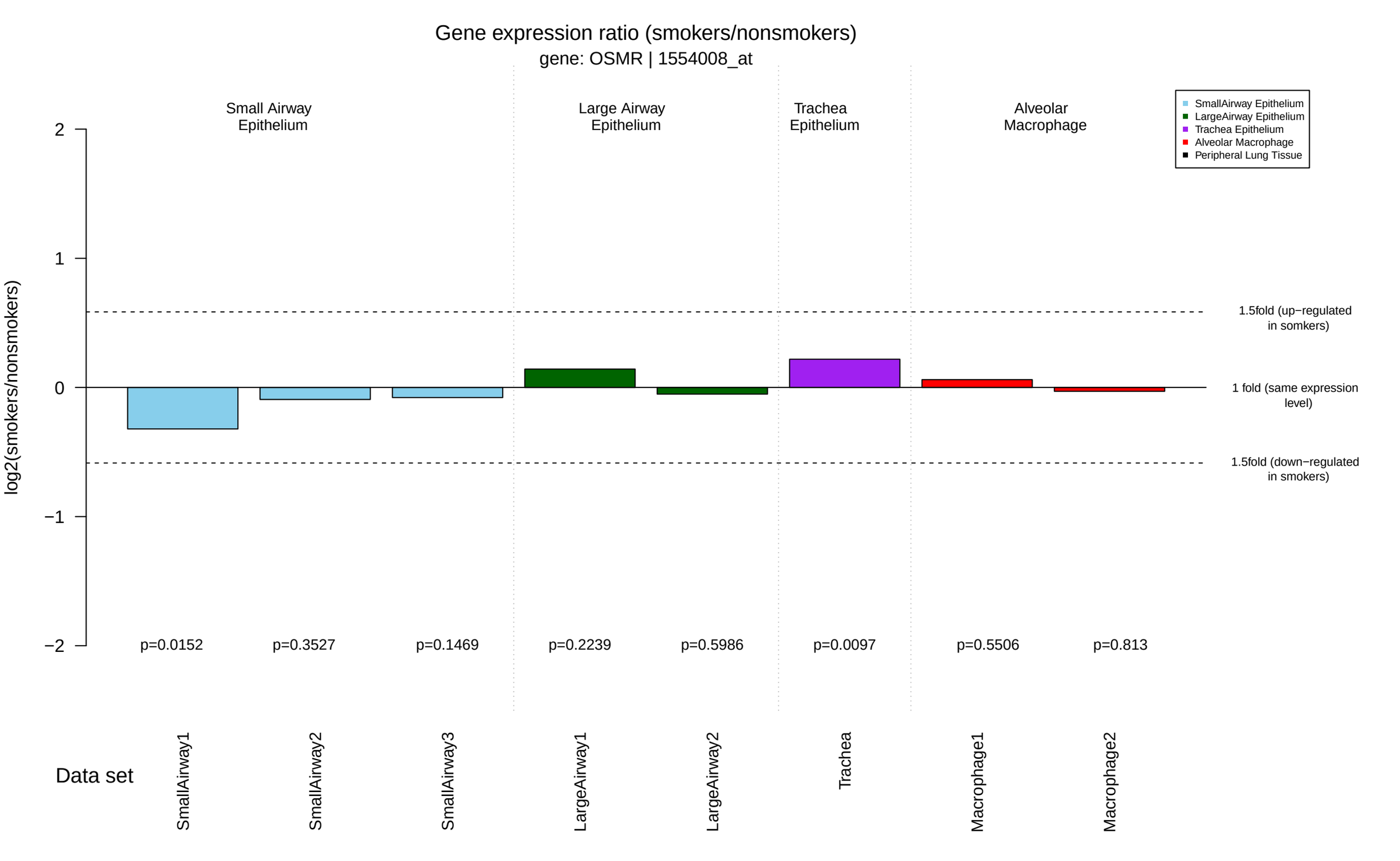
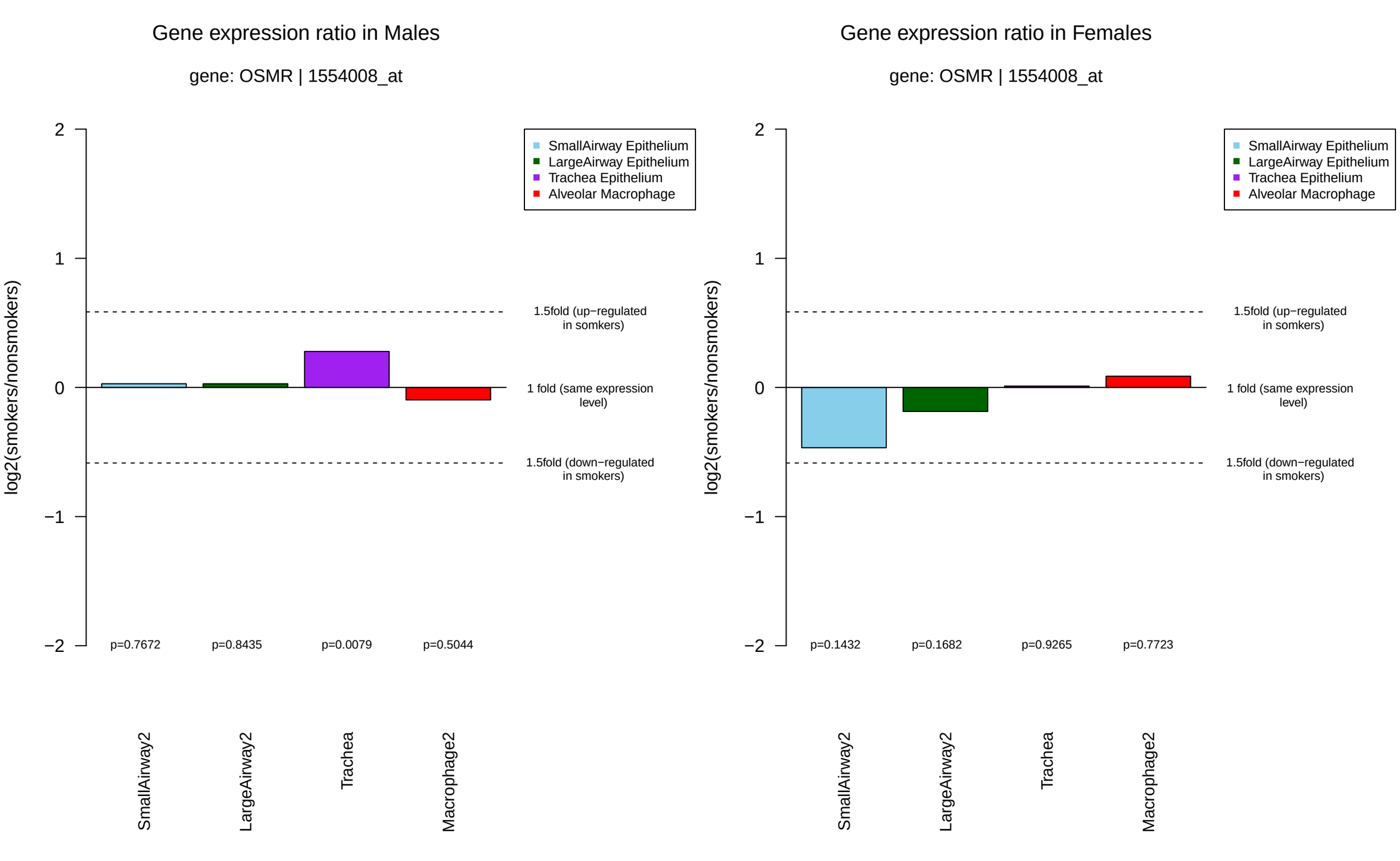
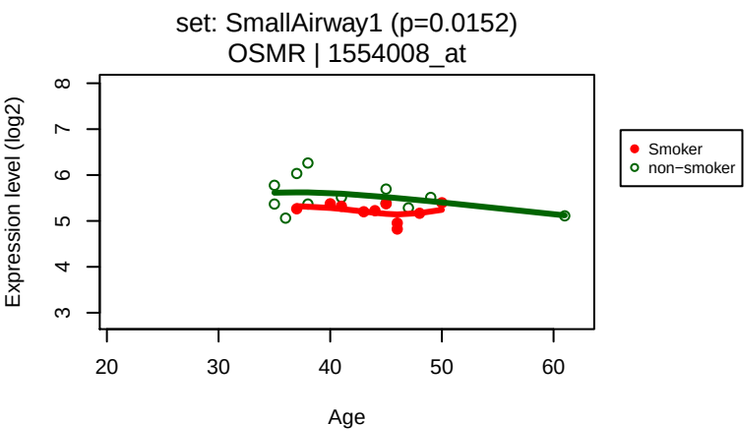
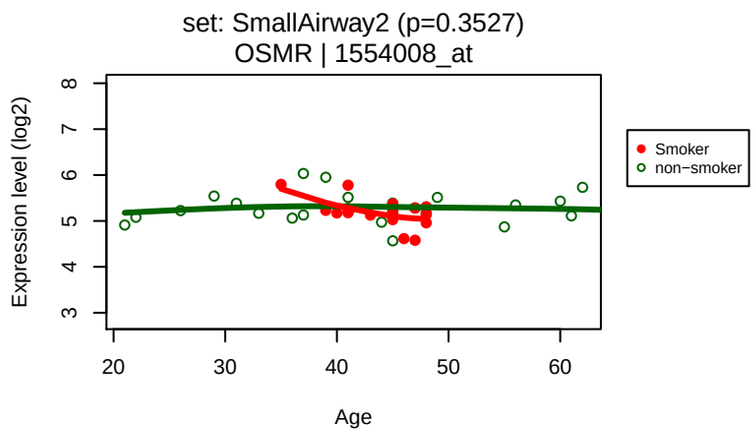
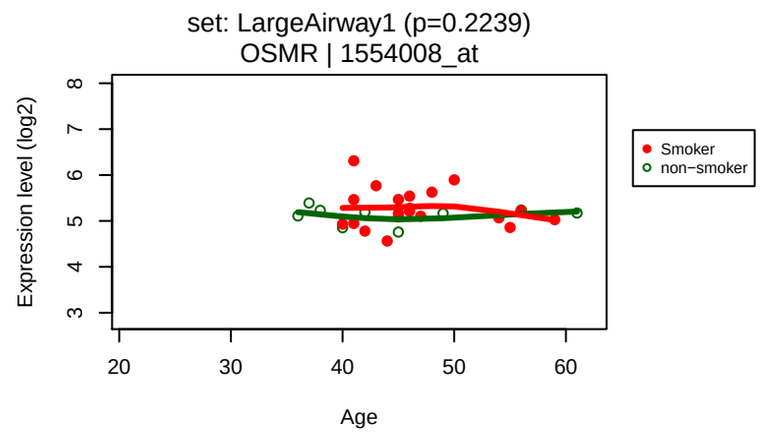

Gene expression of smokers and nonsmokers
gene: OSMR | 1554008_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0152
p=0.3527
p=0.1469
p=0.2239
p=0.5986
p=0.0097
p=0.5506
p=0.813
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

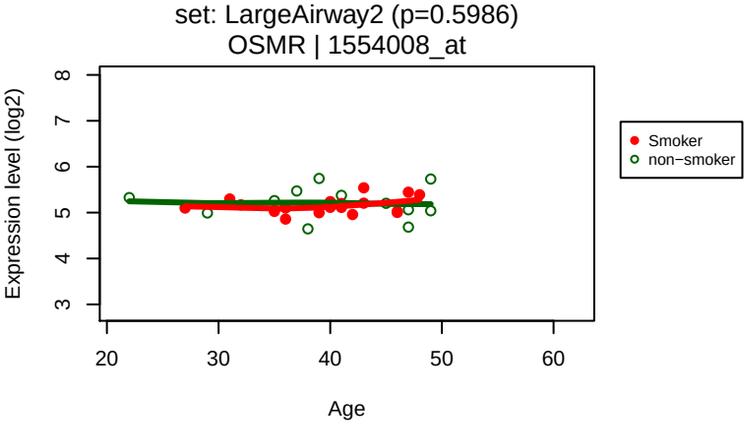

Gene expression ratio (smokers/nonsmokers)
gene: OSMR | 1554008_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0152
p=0.3527
p=0.1469
p=0.2239
p=0.5986
p=0.0097
p=0.5506
p=0.813
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

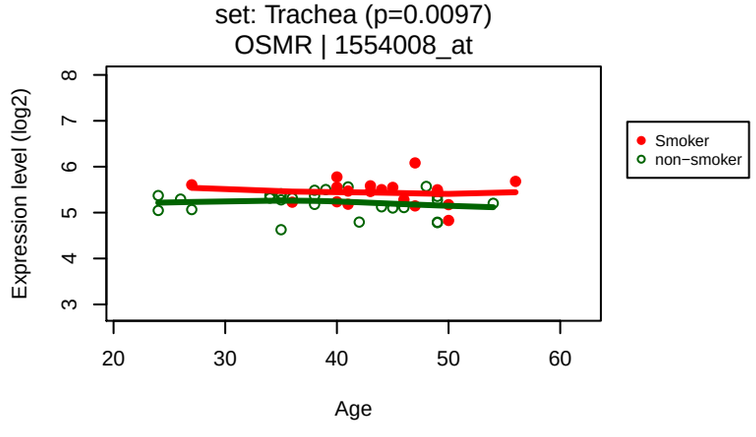

Gene expression ratio in Males
gene: OSMR | 1554008_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7672
p=0.8435
p=0.0079
p=0.5044
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: OSMR | 1554008_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1432
p=0.1682
p=0.9265
p=0.7723
SmallAirway2
LargeAirway2
Trachea
Macrophage2

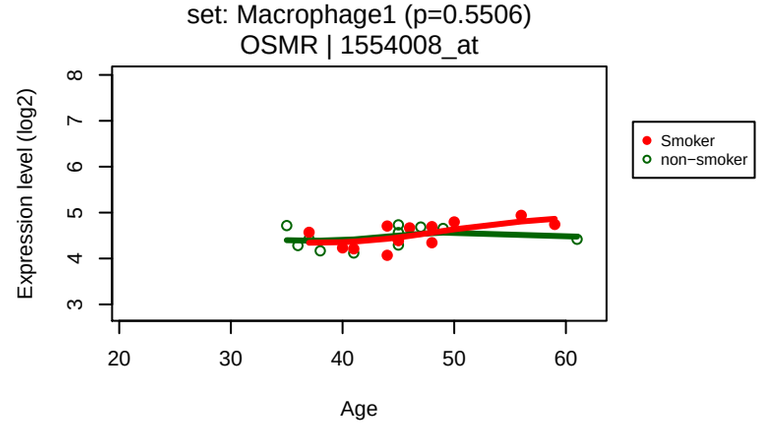

set: SmallAirway1 (p=0.0152)
OSMR | 1554008_at
Expression level (log2)
Age
Smoker
non−smoker

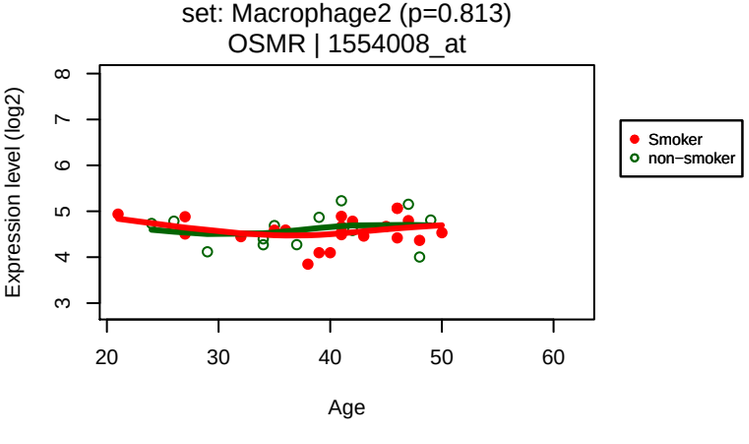

set: SmallAirway2 (p=0.3527)
OSMR | 1554008_at
Expression level (log2)
Age
Smoker
non−smoker

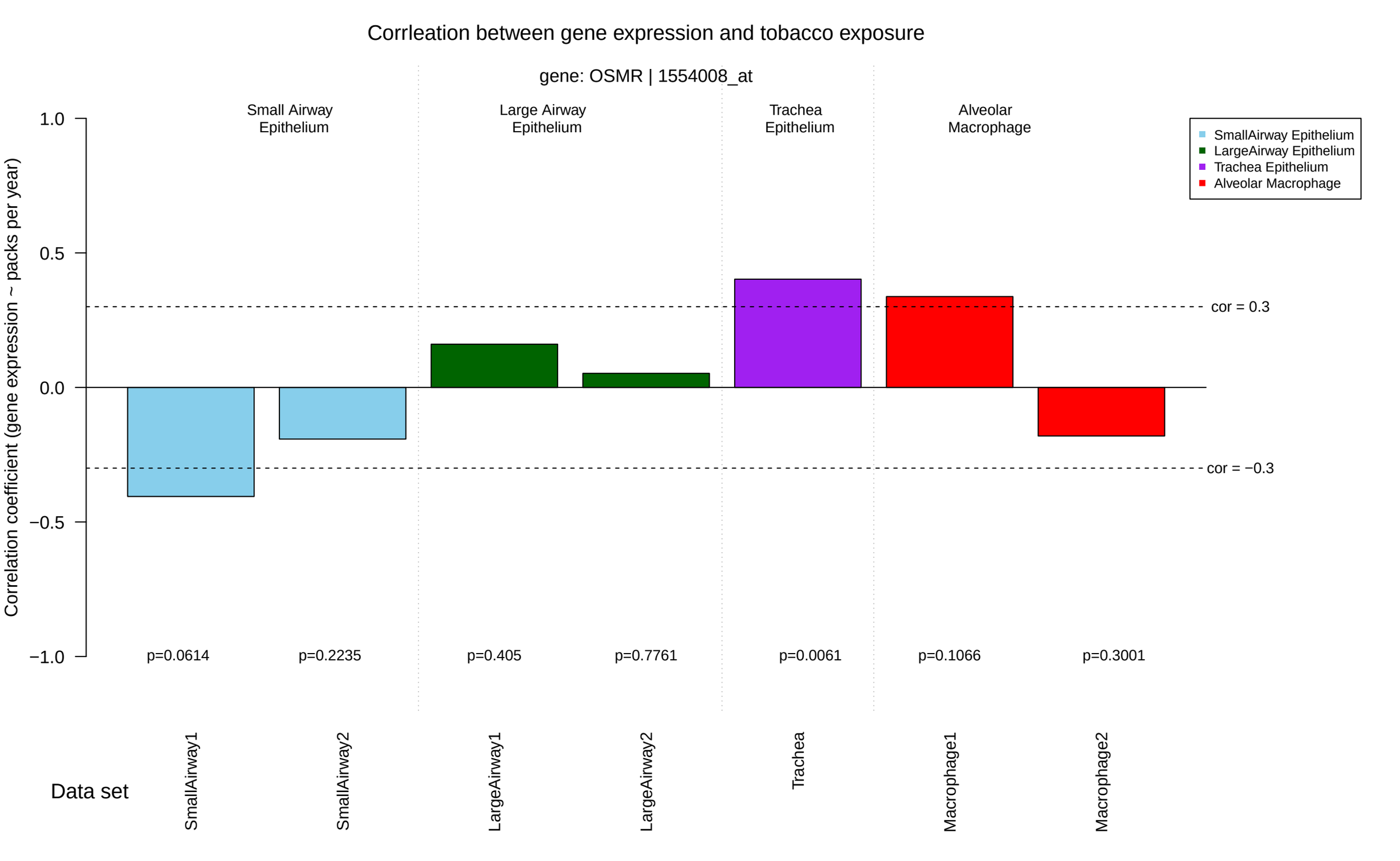

set: LargeAirway1 (p=0.2239)
OSMR | 1554008_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5986)
OSMR | 1554008_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0097)
OSMR | 1554008_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5506)
OSMR | 1554008_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.813)
OSMR | 1554008_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: OSMR | 1554008_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0614
p=0.2235
p=0.405
p=0.7761
p=0.0061
p=0.1066
p=0.3001
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2